# STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-COMPARE-QUASI-COHERENT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2262-2274 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2276-2339, the complete author proof (62 body lines).

*Editorial note.* Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 5 of queue_algebraic-stacks-sheaves.json (pb=3017, stmt 2262-2274, proof 2276-2339).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 070S: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2339–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03DX: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2339–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03FH: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2339–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03DR: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2339–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma equation-restrict: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1547–1549. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-characterize-quasi-coherent-bis: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1819–1836. Cited

*Date*: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-compare: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1430–1452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-compare-locally-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2247–2253. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-compare-morphism: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1553–1570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma remark-compare: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1508–1529. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03DO: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2339–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $S$ be a scheme. Let $\mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids which is representable by an algebraic space $F$. The functor (equation-restrict) defines an equivalence*

$$QCoh(\mathcal{O}_\mathcal{X}) \to QCoh(\mathcal{O}_F), \quad \mathcal{F} \longmapsto \mathcal{F}|_{F_{étale}}$$

*with quasi-inverse given by $\mathcal{G} \mapsto \pi_F^*\mathcal{G}$. This equivalence is compatible with pull-back for morphisms between categories fibred in groupoids representable by algebraic spaces.*

*Proof.* By Lemma lemma-characterize-quasi-coherent-bis we may work with the étale topology. We will use the notation and results of Lemma lemma-compare without further mention. Recall that the restriction functor $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X}) \to Mod(F_{étale}, \mathcal{O}_F)$, $\mathcal{F} \mapsto \mathcal{F}|_{F_{étale}}$ is given by $i_F^*$. By Lemma lemma-compare-locally-quasi-coherent or by Modules on Sites, Lemma 03DO we see that $\mathcal{F}|_{F_{étale}}$ is quasi-coherent if $\mathcal{F}$ is quasi-coherent. Hence we get a functor as indicated in the statement of the lemma and we get a functor $\pi_F^*$ in the opposite direction. Since $\pi_F \circ i_F = \mathrm{id}$ we see that $i_F^*\pi_F^*\mathcal{G} = \mathcal{G}$.

For $\mathcal{F}$ in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ there is a canonical map $\pi_F^*(\mathcal{F}|_{F_{étale}}) \to \mathcal{F}$, namely the map adjoint to the identification $\mathcal{F}|_{F_{étale}} = \pi_{F,*}\mathcal{F}$. We will show that this map is an isomorphism if $\mathcal{F}$ is a quasi-coherent module on $\mathcal{X}$. Choose a scheme $U$ and a surjective étale morphism $U \to F$. Denote $x : U \to \mathcal{X}$ the corresponding object of $\mathcal{X}$ over $U$. It suffices to show that $\pi_F^*(\mathcal{F}|_{F_{étale}}) \to \mathcal{F}$ is an isomorphism after restricting to $\mathcal{X}_{étale}/x = (Sch/U)_{étale}$. Since $U \to F$ is étale, it follows from Remark remark-compare that

$$\pi_F^*(\mathcal{F}|_{F_{étale}})|_{\mathcal{X}_{étale}/x} = \pi_U^*(\mathcal{F}|_{U_{étale}})$$

and that the restriction of the map $\pi_F^*(\mathcal{F}|_{F_{étale}}) \to \mathcal{F}$ to $\mathcal{X}_{étale}/x = (Sch/U)_{étale}$ is equal to the corresponding map $\pi_U^*(\mathcal{F}|_{U_{étale}}) \to \mathcal{F}|_{(Sch/U)_{étale}}$. Since we have seen the result is true for schemes in Descent, Section 03DR[1] we conclude.

Compatibility with pullbacks follows from the fact that the quasi-inverse is given by $\pi_F^*$ and the commutative diagram of ringed topoi in Lemma lemma-compare-morphism. □

[1]Namely, if $U$ is a scheme and $\mathcal{F}$ is quasi-coherent on $(Sch/U)_{étale}$, then $\mathcal{F} = \mathcal{H}^a$ for some quasi-coherent module $\mathcal{H}$ on the scheme $U$ by Descent, Proposition 03DX. In other words, $\mathcal{F} = (\mathrm{id}_{étale,Zar})^*\mathcal{H}$ by Descent, Remark 03FH with notation as in Descent, Lemma 070S. Then we have $\mathrm{id}_{étale,Zar} = \pi_U \circ \mathrm{id}_{small,étale,Zar}$ and hence we see that $\mathcal{F} = \pi_U^*\mathcal{G}$ where $\mathcal{G} = (\mathrm{id}_{small,étale,Zar})^*\mathcal{H}$ is quasi-coherent. Then $\pi_U^* i_U^* \mathcal{F} = \pi_U^* i_U^* \pi_U^* \mathcal{G} = \pi_U^* \mathcal{G} = \mathcal{F}$ as desired.